A Comprehensive Survey on Causal Representation Learning and Causal Discovery in Deep Learning

Abstract

Deep learning has achieved unprecedented success in predictive tasks across diverse domains, primarily by leveraging massive datasets to capture complex statistical associations. However, these models frequently falter when deployed in out-of-distribution (OOD) environments or when subjected to adversarial perturbations, revealing a fundamental over-reliance on spurious correlations rather than true generative mechanisms. To overcome these limitations, the integration of causality into machine learning has emerged as a transformative paradigm. This survey provides an exhaustive examination of two foundational pillars at the intersection of causal inference and deep learning: Causal Discovery and Causal Representation Learning (CRL). While causal discovery seeks to uncover the directed acyclic graphs (DAGs) representing causal relationships among observed variables, CRL addresses the more complex challenge of extracting high-level, causally related latent variables from unstructured, high-dimensional observations such as images, text, and time series. This manuscript systematically reviews the evolution of differentiable causal discovery—from continuous acyclicity constraints to reinforcement learning-based graph search—and thoroughly analyzes the theoretical landscape of identifiability in CRL, encompassing intervention-based, temporal, and multi-view settings. Furthermore, the survey explores the application of causal principles to invariant risk minimization, out-of-distribution generalization, and causal reinforcement learning. By synthesizing algorithmic innovations, theoretical guarantees, and emerging benchmarks, this report outlines the current frontiers and ongoing challenges in constructing truly robust, interpretable, and causally-aware artificial intelligence systems.

Introduction

Statistical machine learning algorithms have achieved state-of-the-art results on benchmark datasets, frequently outperforming humans in pattern recognition and predictive tasks [1, 2, 3]. Despite these triumphs, contemporary deep neural networks remain fundamentally associational; they excel at mapping inputs to outputs based on historical correlations but lack an understanding of the underlying physical or logical mechanisms generating the data. This limitation manifests acutely when models encounter out-of-distribution (OOD) data or unobserved confounders, leading to severe performance degradation [4, 5, 6]. The classic adage that "correlation does not imply causation" is exemplified in deep learning by phenomena such as Simpson's paradox and the exploitation of spurious background features. For example, a vision model might classify a cow in a desert as a camel because the model has inextricably linked the "desert" background to the "camel" label, rather than learning the intrinsic morphological features of the animal [7, 8].

To transcend these limitations, researchers have increasingly turned to the framework of structural causal models (SCMs) and Pearl's Causal Hierarchy, which stratifies causal inference into three distinct levels: Association (seeing), Intervention (doing), and Counterfactuals (imagining) [4, 9, 10]. Traditional machine learning operates almost entirely at the associational level. Elevating artificial intelligence to the interventional and counterfactual levels requires the capability to reason about how a system will behave under exogenous manipulations. Integrating causality can lead to stronger, deeper feature selection and ultimately enable better decision-making in machine learning tasks [10, 11].

This survey focuses on the computational methodologies designed to bridge deep learning and causal inference, broadly categorized into Causal Discovery (CD) and Causal Representation Learning (CRL). Causal discovery algorithms attempt to identify the cause-and-effect relationships among a set of pre-defined, observable variables, typically represented as a Directed Acyclic Graph (DAG) [12, 13, 14]. Historically, causal discovery was dominated by constraint-based and score-based combinatorial search methods, which struggle to scale efficiently with high-dimensional data or capture intricate non-linear relationships [12, 15]. The advent of differentiable causal discovery has revolutionized this space by reformulating combinatorial graph search into continuous optimization problems, allowing for the direct application of deep learning techniques [16, 17, 18].

However, in many modern artificial intelligence applications—such as computer vision, natural language processing, and robotics—the true causal variables are not directly observed [19, 20]. Instead, the system observes raw, high-dimensional, and entangled sensory inputs, such as pixels or audio waveforms. Causal representation learning emerges as the critical bridge, aiming to simultaneously discover the high-level latent causal variables and infer the causal relationships among them [20, 21]. CRL is an inherently ill-posed problem, representing a unification of two notoriously difficult challenges: disentangled representation learning and causal discovery [22, 23]. Consequently, establishing theoretical guarantees for the identifiability of these latent structures—proving that a model can recover the true generative factors up to trivial indeterminacies—represents the most significant ongoing effort in the field [24, 25, 26].

This manuscript provides a comprehensive, rigorous examination of these domains. The subsequent sections define the mathematical foundations of structural causal models, explore the specific algorithmic families driving continuous causal discovery, and dissect the theoretical requirements for identifiability in causal representation learning across various data modalities. Finally, the survey evaluates how these causal representations are utilized in downstream tasks, specifically highlighting invariant risk minimization for OOD generalization and the emerging paradigm of causal reinforcement learning.

Methods

The integration of causal inference with deep learning can be systematically organized into distinct methodological families. This section delineates the evolution from traditional causal discovery to continuous optimization frameworks, followed by the theoretical and algorithmic foundations of causal representation learning, and concluding with the application of causal principles to generalization and reinforcement learning.

Continuous and Differentiable Causal Discovery

Causal inference relies on a formal mathematical language to describe the interactions between observed variables. A Structural Causal Model (SCM) provides this formalism, defining a system of variables and the mechanisms that dictate their values [12, 16]. Let a system be described by a d-dimensional random vector $X = [X_1, X_2, \dots, X_d]$. An SCM asserts that each variable X_j is computed as a deterministic function of its direct causes and an unobserved, exogenous noise term E_j. Formally, this is written as $X_j = f_j(Pa(X_j), E_j)$, where $Pa(X_j)$ denotes the parent set of X_j, and f_j represents the structural causal mechanism [13, 17]. The structure of these relationships is universally encoded using a Directed Acyclic Graph (DAG), denoted as $G = (V, E)$, where the vertices V correspond to the variables in X, and the directed edges E indicate direct causal influence [18, 27].

Traditional approaches to causal discovery fall into two primary categories. Constraint-based methods, such as the Peter-Clark (PC) algorithm and the Fast Causal Inference (FCI) algorithm, utilize statistical conditional independence tests to eliminate edges and orient the remaining connections, yielding a Markov Equivalence Class of valid DAGs [14, 28]. Score-based methods, such as Greedy Equivalence Search (GES), assign a scoring metric, such as the Bayesian Information Criterion (BIC), to candidate DAGs and employ heuristic search algorithms to navigate the discrete space of possible graphs [14, 15]. The primary limitation of these traditional frameworks is the combinatorial explosion of the search space, as the number of possible DAGs grows super-exponentially with the number of variables d, rendering exhaustive search computationally intractable [17, 28].

The NOTEARS Paradigm and Trace Exponential Constraints

The seminal breakthrough in continuous causal discovery was the introduction of the NOTEARS (Non-combinatorial Optimization via Trace Exponential and Augmented lagRangian for Structure learning) algorithm [16, 17]. NOTEARS revolutionized the field by introducing a smooth algebraic characterization of acyclicity. Assuming a linear SCM where $X = XW + E$, the graph is encoded by the weighted adjacency matrix $W \in \mathbb{R}^{d \times d}$.

The NOTEARS framework establishes that the matrix W corresponds to a DAG if and only if the following trace exponential constraint is satisfied:

$$h(W) = \mathrm{tr}(e^{W \circ W}) - d = 0$$

where $\circ$ denotes the Hadamard (element-wise) product, and e^M is the matrix exponential [16, 17]. Because the trace of the matrix exponential equals the number of nodes d strictly when the graph contains no cycles, this equality constraint exactly characterizes the space of DAGs [17]. Causal discovery is thus recast as minimizing a loss function subject to $h(W) = 0$, typically solved using augmented Lagrangian methods.

Non-linear and Neural Structural Causal Models

While the original NOTEARS framework was constrained to linear relationships, the continuous formulation naturally paved the way for deep learning architectures capable of capturing non-linear mechanisms. GraN-DAG (Gradient-based Neural DAG Learning) extended the continuous constrained optimization framework by parametrizing the conditional distributions of each variable using multi-layer perceptrons (MLPs) [29, 30]. In GraN-DAG, each neural network models the non-linear relationships between a node and all potential parents. By applying a modified acyclicity constraint that operates on the neural network paths rather than simple linear coefficients, GraN-DAG successfully captures complex, non-linear generative processes [29, 30].

Concurrently, the DAG-GNN architecture proposed utilizing Graph Neural Networks (GNNs) within a Variational Autoencoder (VAE) framework to learn the causal graph [31, 32]. DAG-GNN models the adjacency matrix explicitly as a learnable parameter within a graph convolutional structure. The encoder maps the observed data X to latent representations Z, and the decoder reconstructs X by applying the learned adjacency matrix. By optimizing the Evidence Lower Bound (ELBO) alongside a structural acyclicity penalty, DAG-GNN effectively handles both continuous and discrete variables [31, 32].

Advanced Acyclicity Characterizations: DAGMA and Differentiable d-Separation

Despite the success of the trace exponential constraint, it poses numerical stability issues and computational bottlenecks for large graphs due to the calculation of the matrix exponential, which is prone to ill-conditioning [16, 33]. The DAGMA (Learning DAGs via M-matrices and a Log-Determinant Acyclicity Characterization) algorithm proposed an alternative formulation, introducing a log-determinant function to characterize acyclicity:

$$h(W) = -\log\det(sI - W \circ W) + d\log s = 0$$

where s is a scalar parameter [33, 34]. This formulation leverages the properties of M-matrices, demonstrating faster convergence and superior scalability compared to NOTEARS-based methods [33, 34]. Recent literature has also explored integrating the robust principles of constraint-based discovery with continuous optimization. The DAGPA method introduces differentiable functions of d-separation and d-connection, framing conditional independence scoring through a continuous relaxation based on percolation theory and soft logic [14, 35].

Algorithm Family	Representative Methods	Core Innovation	Functional Assumption

Linear Continuous	NOTEARS, GOLEM	Recasting DAG search as a constrained continuous optimization problem using trace exponential matrices.	Linear Structural Equation Models (LSEMs)
Neural/Non-linear Continuous	GraN-DAG, NOTEARS-MLP	Employing Multi-Layer Perceptrons to capture non-linear causal relations while maintaining continuous structural penalties.	Non-linear additive noise models
Graph Generative Models	DAG-GNN, CausalVAE	Using Variational Autoencoders and Graph Convolutional Networks to map and optimize structural adjacency matrices.	Generalized deep non-linear functions
Log-Determinant Optimizers	DAGMA, Dagma-DCE	Replacing matrix exponential with a more stable log-determinant constraint derived from M-matrices.	Model-agnostic / Flexible

Reinforcement Learning and Bayesian Optimization in Graph Search

An alternative to strict analytical constraints involves formulating DAG discovery as a sequential decision-making problem. Methods like CORL (Causal discovery with Ordering-based Reinforcement Learning) utilize modern RL architectures to search the reduced space of variable orderings, subsequently pruning edges to guarantee acyclicity [12, 36]. Expanding on this, the ALIAS (reinforced dAg Learning wIthout Acyclicity conStraints) framework constructs a surjective parametrization that maps any continuous vector space directly into the space of all valid DAGs, bypassing explicit acyclicity regularizers [37]. Similarly, the DrBO (DAG recovery via Bayesian Optimization) algorithm utilizes dropout neural networks as a surrogate model to continually guide the Bayesian exploration of the DAG space, vastly improving sample efficiency [38, 39].

Causal Representation Learning (CRL)

While causal discovery assumes the causal variables themselves are known and observed, this assumption fails for most unstructured data processed by modern AI (e.g., images, video, text). Causal Representation Learning (CRL) seeks to solve this by simultaneously recovering the high-level latent variables Z and their causal relationships from the low-level, entangled observations X [20, 21, 23].

Formally, CRL considers a data-generating process where causally related latent variables $Z = \{Z_1, \dots, Z_k\}$ are mapped to high-dimensional observations $X \in \mathbb{R}^d$ via an unknown, generally non-linear mixing function g, such that $X = g(Z)$ [21, 40]. The objective is to invert this process, learning an encoder that isolates the true components of Z and recovers their underlying DAG structure.

The central theoretical challenge in CRL is *identifiability*. Without stringent inductive biases or structural assumptions, non-linear representation learning is fundamentally unidentifiable; there exist infinitely many spurious latent representations and mixing functions that can perfectly reconstruct the observational data distribution while completely misrepresenting the true latent causal structure [24, 25, 41]. This pursuit is closely linked to Non-linear Independent Component Analysis (ICA). While standard non-linear ICA assumes the latent sources are statistically independent, CRL faces a strictly harder problem because the latent variables are, by definition, causally dependent on one another [41, 42].

Identifiability via Interventions

One of the most robust mechanisms to guarantee identifiability is leveraging interventional data [20, 26, 40]. When a system undergoes localized, sparse changes—interventions—the resulting shifts in the data distribution carry precise structural information about the causal mechanisms. Theoretical analysis indicates that observing the system under a sufficient number of distinct interventions can fully identify the latent SCM. Score-based CRL algorithms exploit this by computing the score functions (the gradients of the log-density of the distributions, $\nabla \log p(x)$) across different interventional environments. Because interventions create sparse variations in the causal mechanisms, aligning these score variations enables the exact inversion of the non-linear mixing function g [40, 43].

Recent literature has mapped the identifiability landscape based on the nature of the interventions. If the mixing function is linear but the causal mechanisms are non-linear, a single hard (perfect) intervention per node is sufficient to perfectly recover the latent DAG and the variables up to scaling [40, 44]. Remarkably, even when the specific targets of the interventions are unknown (uncoupled environments), identifiability holds [44, 45]. For general non-linear transformations, two stochastic hard interventions per node are theoretically sufficient [40]. While the majority of these proofs apply to infinite-sample asymptotic regimes, recent advances have established finite-sample, non-asymptotic identifiability guarantees, proving that causal representations can be learned reliably even with limited data [26, 46].

Temporal and Dynamical Causal Representation Learning

In the absence of interventional data, temporal structures provide a powerful mechanism to achieve identifiability. Time-series data implicitly encode causal directionality, which serves as a natural constraint on the representation space [47, 48]. Models like the Time-Causal Variational Autoencoder (TC-VAE) enforce temporal causality by strictly constraining the encoder and decoder architectures such that latent variables at time t depend solely on historical data, utilizing causal Wasserstein distances to enforce transport constraints [47, 49].
Furthermore, the Causally Hierarchical Latent Dynamics (CHiLD) framework addresses the complex reality that time-series data operate across multiple levels of temporal abstraction [50, 51]. Standard temporal CRL methods fail to capture multi-layer latent dynamics from single-timestep observations. The CHiLD framework proves mathematically that the joint distribution of hierarchical latent variables is completely and uniquely identifiable if the model utilizes three conditionally independent sequential observations (a 3-measurement model) alongside the natural sparsity of the hierarchical graph [50, 52]. By employing temporal contextual encoders and normalizing flow-based prior networks, CHiLD successfully disentangles non-stationary, multi-layer latent variables [50, 51]. Similarly, handling non-stationary transitions—where the rules governing the causal dynamics change over time or across domains—is a paramount challenge addressed by architectures like CtrlNS (Causal Temporal Representation Learning with Nonstationary Sparse Transition), which clusters transitions and enforces sparsity on the variations between domains [53, 54].

Multi-View and Multimodal CRL

In settings lacking explicit interventions or temporal sequences, multimodal or multi-view data can establish identifiability. In real-world environments, a single underlying causal state often generates multiple distinct types of observations [24, 55]. Theoretical frameworks for multimodal CRL demonstrate that when observed variables are non-linear functions of partially shared subsets of latent variables across different views, the shared causal representation can be identified [24, 55]. Using contrastive learning paradigms across these multiple partial views restricts the possible space of the mixing function, ensuring that the information shared across modalities represents the true underlying causal concepts [22, 55].

Domain Condition	Identifiability Mechanism	Key Architectures / Frameworks
Interventional Data	Alignment of score functions across distinct	Score-based CRL, IL-SCM

	interventional environments.	
Temporal Data (Stationary)	Time-delayed causal priors and sequence transition sparsity constraints.	TC-VAE, TDRL
Temporal (Hierarchical)	Contextual encoding over 3-measurement conditional independent blocks.	CHiLD
Multimodal / Multi-view	Contrastive mapping of partially shared subsets across distinct data modalities.	Multi-View CRL, CausalVAE
Discrete Data Domains	Bipartite generative measurements with sequential Monte Carlo posterior mapping.	DCRL

Causality for Out-of-Distribution Generalization

The representations and graphs extracted via causal discovery and CRL are fundamental to solving robust decision-making and generalization challenges in downstream deep learning tasks. A persistent vulnerability in empirical risk minimization (ERM) is its tendency to exploit spurious correlations that hold in the training data but fail in the real world [56, 57].
Invariant Risk Minimization (IRM) was proposed as a paradigm to estimate invariant, causal correlations across multiple training environments [56, 58]. The core tenet of IRM is to learn a data representation such that the optimal linear classifier operating on top of that representation is identical and simultaneously optimal across all observed environments [58, 59]. By enforcing this environmental invariance, IRM mathematically penalizes the model for relying on spurious features, pushing it to latch onto the true causal mechanisms that generated the label [58, 60].
While theoretically elegant, the practical implementation of IRM (IRMv1) has exposed several failure modes. In settings where spurious features exhibit strong correlations or where the problem is highly non-linear, IRM can fail to capture the true invariance, occasionally performing worse than standard ERM [59, 61]. To address these shortcomings, researchers have developed multiple extensions. Risk Extrapolation (REx) proposes a robust optimization penalty over the variance of training risks across environments, specifically targeting the magnification of extreme distributional shifts [62, 63]. Other frameworks, such as Sparse Invariant Risk Minimization (SparseIRM), enforce global sparsity constraints during training to prevent the network from memorizing spurious features through overparameterization [64]. Furthermore,

methods like Multi-Expert Distributionally Robust Optimization (MEDRO) explicitly expand the uncertainty set to capture fine-grained subpopulation shifts by deploying environment-specific expert heads over a shared causal feature extractor [65]. In the graph domain, methods like PrunE target spurious correlations in graph neural networks by actively pruning structurally irrelevant edges, preserving only the causally invariant subgraphs necessary for robust prediction [66].

Causal Reinforcement Learning

The principles of causality are increasingly being integrated into Reinforcement Learning (RL), yielding a subfield known as Causal Reinforcement Learning (CRL) [67, 68]. Standard RL agents operate entirely through trial-and-error, lacking common knowledge of the environment and struggling to adapt to structural changes [67, 69].

Causal RL bridges this gap by explicitly acknowledging that the environments in which an RL agent operates are governed by structural causal models [68, 70]. The framework mathematically unites RL with Pearl's causal hierarchy, enabling agents to leverage structural invariances and formal counterfactual logic [67, 70]. By equipping agents with a causal world model, Causal RL addresses numerous complex tasks, including generalized policy learning (combining off-policy observational data with targeted online interventions), causal imitation learning (learning optimal behavior from an expert in the presence of unobserved confounders), and counterfactual decision making [70, 71]. Causal models allow the agent to evaluate counterfactual scenarios, drastically improving sample efficiency, safety, and the ability to generalize policies to novel, unencountered environments [67, 72].

Challenges and Prospects

Despite rapid theoretical advancements in causal representation learning and differentiable causal discovery, the transition to robust, scalable, real-world systems faces several profound challenges.

Evaluation and Benchmarking

A critical bottleneck in the field is the difficulty of evaluating causal algorithms, owing to the absence of accessible ground-truth causal structures in real-world high-dimensional data [73, 74]. Historically, models have been evaluated on synthetic, low-dimensional datasets or highly stylized image datasets, which fail to reflect the complexity of physical environments [73, 75]. To evaluate algorithm scalability accurately, the community has introduced complex environments such as CausalVerse. CausalVerse provides a high-fidelity simulated visual dataset comprising over 200,000 images and 300 million video frames across varied domains (static generation, robotic manipulation, traffic physics) where the exact ground-truth causal graphs and latent variables are entirely known by the simulator [73]. Similarly, the TCD-Arena provides a highly modular testing kit to evaluate time-series causal discovery algorithms against dozens of explicit assumption violations, revealing the nuanced failure modes of contemporary algorithms [76]. Furthermore, researchers are continuously questioning the validity of standard metrics like the Structural Hamming Distance (SHD), noting that continuous optimization techniques can sometimes exhibit excellent SHD scores while returning overly sparse graphs with poor true positive rates, signaling optimization collapse [77].

Assumption Violations and Scalability

Almost all continuous causal discovery and CRL models rely on foundational assumptions—such as non-parametric additive noise, linearity of the mixing function, or the absence of unobserved confounders—that are frequently violated in real-world datasets [13, 78]. While models attempt to account for heteroscedasticity and confounding, scaling these exact constraint-based optimizations to thousands of variables while maintaining gradient stability remains an open systems engineering challenge [13, 79]. Moreover, as data privacy constraints tighten, Federated Causal Discovery (FCD) has emerged to enable collaborative structural learning without centralizing raw data. In FCD, local clients compute structural properties and securely aggregate them on a central server, introducing further complexities regarding network heterogeneity and mechanism identifiability across decentralized nodes [80, 81].

Generative AI and Large Language Models

Looking forward, the intersection of Large Language Models (LLMs) and causal representation learning holds massive potential. Existing generative models are highly adept at identifying linguistic and visual dependencies but fundamentally lack the ability to manipulate causal relationships reliably, often leading to hallucinations [82, 83]. Introducing identifiable temporal causal representation frameworks directly into the high-dimensional concept space of LLMs is an active area of mechanistic interpretability research [83, 84]. By factorizing contextual embeddings into causal factors (sufficient for prediction) and non-causal factors (redundant features), future models aim to inherently resist spurious reasoning, effectively serving as robust causal simulators for human-aligned planning tasks [84, 85].

Conclusion

The fusion of deep learning with causal inference represents one of the most critical evolutions in modern artificial intelligence. As demonstrated throughout this survey, differentiable causal discovery has successfully shattered the historical combinatorial constraints of graph search, allowing deep neural architectures to optimize complex topological structures natively via smooth continuous constraints. Concurrently, causal representation learning has laid the mathematical groundwork for discovering these structures directly from entangled, raw sensory data. Through the rigorous application of interventional diversity, temporal dynamics, and multi-view contrastive learning, the field has proven that the recovery of latent causal variables is fundamentally identifiable under the right conditions. When these structural causal models are integrated into downstream systems—whether through invariant risk minimization to ensure robust out-of-distribution generalization, or through causal reinforcement learning to build highly adaptive, counterfactual-reasoning agents—the result is a dramatic shift away from brittle, associational curve-fitting. As the community continues to develop higher-fidelity benchmarks and tackles the nuances of non-stationary, heteroscedastic real-world environments, causally-aware deep learning stands positioned to deliver the next generation of safe, interpretable, and genuinely intelligent autonomous systems.

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